

1 Controlled intervention

Reference branch

ERA5 state
 $x(t)$

unchanged input
 $x^{(0)} = x$

Patched branch

WB2 climatology
 $c_v(d, h)$
same season + UTC

$x_v^{(\alpha)} = (1 - \alpha)x_v + \alpha c_v$
 $\alpha = 0, .2, \dots, 1$
date and other 18 fields fixed

2 Matched rollouts

Baseline

f_θ
frozen weights

$\hat{y}_u^{(0)}$
 $+6h, +24h$

Patched

f_θ
same weights

$\hat{y}_u^{(v, \alpha)}$
 $+6h, +24h$

same date, lead τ , and model parameters θ

PAIR

4 Evidence products

1. Rank cross-field transfer
 2. Test field loss weights
 3. Test latitude weights
- Confirm by retraining

Directed transfer
matrices
 19×19

Dose-response curves
 $\alpha = 0, .2, \dots, 1$

Error geography
maps + latitude profiles

OUT

repeat over dates, leads, doses, fields, and grid cells

3 Paired diagnostics

S_{vu}
forecast displacement
patched vs. baseline

ΔACC_{vu}
verified skill change
against ERA5 targets

POOL

SCORE

area-weighted and evaluated separately for each input v and output u